

Standard Resistor Values: E3, E6, E12, E24, E48 & E-Series

stand the E3, E6, E12, E24, E48 & E96 series of standard resistor values: why they are, tables of values, why they are used, how they were derived, etc. and about the summary infographic.

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Resistor values are organised into a set of different series of preferred values or standard resistor values.

Standard resistor values have a logarithmically based sequence and this enables the different values to be chosen in such a way that they relate to the component tolerance or accuracy.



Tolerances are generally $\pm 20\%$, $\pm 10\%$, $\pm 5\%$, $\pm 2\%$ and $\pm 1\%$. More accurate tolerances are available for some resistors, but these are not as widely available and costs are higher.

By using these standard resistor values, from a variety of manufacturers can be chosen, making sourcing much easier and the cost of the components much less.

Similarly the same approach and sets of value ranges are used by a number of other electronic components such as capacitors, Zener diodes, inductors, etc.

By using the more commonly used values in any electronic circuit design, not only are they more easily available from electronic component distributors, but costs can be reduced because the number of component types on any circuit board or other assembly can be reduced, and this makes significant cost savings in manufacture.



SMD Resistors on a PCB along with other components
Even SMD components follow the E-Series values

Standard resistor values

Standard resistor values are organised into a set of series of values known as the E-series. The different values are chosen such that the top of the tolerance band of one value and the bottom of the tolerance band of the next one overlap.

The E-series is built on a logarithmically based sequence. This ensures that the values are spaced out according to a component's tolerance or accuracy.

The goal is to ensure that the "tolerance spread" of one value perfectly meets the spread of the next. For example:

E3 series: Designed for a $\pm 10\%$ tolerance.

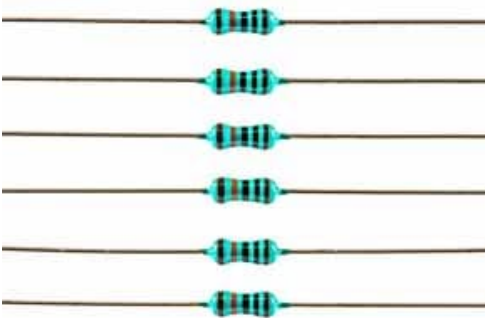
E6 series: Designed for a $\pm 5\%$ tolerance.

Because of this spacing, no matter what exact resistance your circuit requires, a standard E-series resistor will be "close enough" to fulfill the requirement within its specified error margin.

For example a resistor that has a value of 1 ohm and a tolerance of $\pm 20\%$. The actual resistance at the top of its tolerance band is 1.2 ohms. Take then a resistor with a value of 1.5 ohms. The resistance of this component at the bottom of its tolerance band is 1.2 ohms. This process is followed through for all the values in a decade, creating standard resistor values for each tolerance.

Different sets of standard resistor values are known by their E-series numbers: E3 has three resistors in each decade, E6 has six, E12 has twelve, and so forth.

The most basic series within the E range is the E3 series which has just three values: 1, 2.2 and 4.7. This is seldom used because the associated tolerance is too wide for most of today's applications, although the basic values may be used more widely to reduce stock holding.



Leaded metal film resistors

The E6 series has six values in each decade for a $\pm 20\%$ tolerance, E12 series with 12 values in each decade for a $\pm 10\%$, E24 series with 24 values in each decade for a $\pm 5\%$ tolerance. Values for resistors in these series are given below. Further series (E48 and E96) are available, but are not as common as the ones given below.

nd E12 resistors are available in virtually all types of resistor. However the E24 series, being a much closer series is only available in the higher tolerance types.

Resistors

I film resistors that are in common use today as well as metal oxide film resistors are available in the E24 are several other types. Carbon types are rarely available these days and in any case would only available er tolerance ranges as their values cannot be guaranteed to such a close tolerance.

ries preferred or standard resistor value ranges are internationally accepted and have been adopted by ral standards organisations. The EIA (Electrical Industries Association) which is based in North America is isation that has adopted the system and as a result the resistor value series are often referred to as the EIA resistor values.

SUMMARY OF EIA PREFERRED OR STANDARD Resistor VALUE SERIES		
S	TOLERANCE (SIG FIGS)	NUMBER OF VALUES IN EACH DECADE
	>20%	3
	20%	6
	10%	12
	5%	24
	[normally also available in 2% tolerance]	
	2%	48
	1%	96
	0.5%, 0.25% and higher tolerances	192

æ metal film [resistors](#) now widely used for axial resistors and the surface mount resistors are normally in 1% and 2% tolerance ratings even when included in the E24, E12, E6 and E3 ranges.

ies values are split into two groups which have slightly different numbering, although they follow the same erology:

E24: For this lower end of the E series used for resistors, capacitors and other component values, the lfference is that the numbers have only two significant figures as this is all that is really needed

E192: For the E48 to E192 series there significant figures are used for all the values as it is necessary to d them more accurately in view of the higher number of values needed.

seen that some of the values in the E24 series do not exist in E48 to E192 series. This arises dues to the ounding rules used.

Mathematics

rrred and standard values of other components

æm for adopting standard electronic component values works very well for resistors. It is also equally r for other electronic components. The same concept of using values in a standard list that are determined rance of the components is equally applicable.

ies is also used for capacitors, inductors and a number or some other electronic components and it applies aded and surface mount devices.



for capacitors some of the lower order series are used - E3, E6 as the values on many capacitors do not gh tolerance. Electrolytic capacitors typically have a very wide tolerance, although others such as many ypes have a much tighter tolerance and many be available in ranges conforming to the E12 or even E24

Example of electronic components that follow the EIA E series preferred values is Zener diodes for their nominal voltages. The Zener diode standard voltages typically conform to the E12 values although E24 series values are also available - especially 5.1 volt Zener diode for 5 volt rails. Again, this applies to both styles of component: leaded and surface mount devices.

For other electronic components, opting for Zener diode values in the E12 rather than E24 range will generally mean that the values are more widely available from sources like electronic component distributors. That said the 5.1 volt Zener diodes that have an E24 value are commonly used as this corresponds to the standard logic supply rail.

Standard Resistor E series

Preferred values or standard resistor values can be summarised in a tabular form to give the different values for each decade.

Resistor technology enables very close tolerance levels to be achieved, yet there is still a great benefit in using standard resistor values even from the E3 series.

Resistors

By using standard resistors from the E3 series or even E6, this reduces the number of different types of resistor used in a design and this considerably simplifies purchasing and manufacturing processes as well as reducing costs. Often it is necessary to keep to the E3 or E6 standard resistor values, only using those in the E12, E24, E48 or E96 if it is absolutely necessary.

An example where values can be kept within the E3 series occurs with digital design where a pull up or pull down resistor is needed. The exact value is of little consequence - only a value within the approximate region is needed. In such cases, for resistors the value can be selected within the E3 series.

Where more analogue designs a little more flexibility is often needed, but even E6 or E12 standard resistor values can be used. It is difficult in most electronic circuit designs. Occasionally E24, E48, E96 or even E192 series values may be required for high accuracy and close tolerance requirements: filters, oscillators, measurement applications, etc.

Standard Resistor E series tables of values

Below are the common resistor values used in electronic circuit designs. They are the standard E3, E6, E12, E24, E48 and E96 resistor values.

E3 STANDARD RESISTOR SERIES

1.0	2.2	4.7
-----	-----	-----

E3 series resistors are the most widely used and hence these values will be the most common resistor values in the electronics industry. They are particularly useful for resistor values that are not in any way critical. By using this series, the number of different components in any electronic circuit design can be reduced and this reduces manufacturing costs by reducing inventory and the additional management and set up required for component types in a design.

E6 STANDARD RESISTOR SERIES

1.0	1.5	2.2
3.3	4.7	6.8

E12 series resistor values are also widely used within the industry. They provide a wider range of common resistor values that can be used in electronic circuit designs and this can be essential for many analogue designs.

E12 STANDARD RESISTOR SERIES

1.0	1.2	1.5
1.8	2.2	2.7
3.3	3.9	4.7
5.6	6.8	8.2

E24 STANDARD RESISTOR SERIES

1.0	1.1	1.2
1.3	1.5	1.6

E24 STANDARD RESISTOR SERIES

1.8	2.0	2.2
2.4	2.7	3.0
3.3	3.6	3.9
4.3	4.7	5.1
5.6	6.2	6.8
7.5	8.2	9.1

in the E48 series tend to be used for high tolerance applications. They are much less widely used than E24 only those electronic components in the E12 or E6 series. They tend to be used for circuit designs where a precision resistor may be needed to give a close tolerance output, or for measurement, etc.

Resistors

E48 STANDARD RESISTOR SERIES

1.00	1.05	1.10
1.15	1.21	1.27
1.33	1.40	1.47
1.54	1.62	1.69
1.78	1.87	1.96
2.05	2.15	2.26
2.37	2.49	2.61
2.74	2.87	3.01
3.16	3.32	3.48
3.65	3.83	4.02
4.22	4.42	4.64
4.87	5.11	5.36
5.62	5.90	6.19
6.49	6.81	7.15
7.50	7.87	8.25
8.66	9.09	9.53

and indeed, other forms of electronic component within the E96 series are rarely used, although they are used in certain electronic designs for very close tolerance requirements. Being less commonly used, they may not be readily available through electronic component distributors and they are likely to be more expensive, although it should not be too much difficulty in obtaining them.

E96 STANDARD RESISTOR SERIES

1.00	1.02	1.05
1.07	1.10	1.13
1.15	1.18	1.21
1.24	1.27	1.30
1.33	1.37	1.40
1.43	1.47	1.50
1.54	1.58	1.62
1.65	1.69	1.74
1.78	1.82	1.87
1.91	1.96	2.00
2.05	2.10	2.16
2.21	2.26	2.32
2.37	2.43	2.49
2.55	2.61	2.67
2.74	2.80	2.87
2.94	3.01	3.09
3.16	3.24	3.32
3.40	3.48	3.57
3.65	3.74	3.83
3.92	4.02	4.12
4.22	4.32	4.42
4.53	4.64	4.75

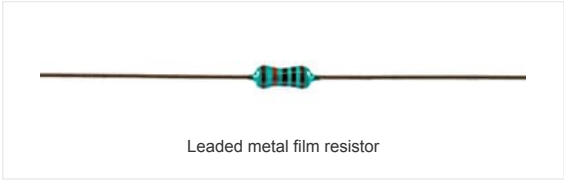
E96 STANDARD RESISTOR SERIES		
4.87	4.99	5.11
5.23	5.36	5.49
5.62	5.76	5.90
6.04	6.19	6.34
6.49	6.65	6.81
6.98	7.15	7.32
7.50	7.68	7.87
8.06	8.25	8.45
8.66	8.87	9.09
9.31	9.53	9.76

! series of standard resistor values also exist, but their use is much smaller than those in the other ranges we. Their tolerance is either 0.5 or 0.25% and this results in increased costs along with the fact that there is gher number of [resistors](#) in the range.

resistors in ranges up to E24 are widely available, it often helps in any design to focus on using as small a f resistor values as possible. This will reduce the number of different components in a design and for large duction, this will help reduce costs.

lopment of E-series values

ly days of radio and electronics, in the first half of the twentieth century, there was little or no standardisation The values chosen for [electronic components](#) like resistors and capacitors were determined by different rers.



ed a number of difficulties for electronic circuit design because often the supplier needed to be identified so lectronic component value could then be chosen.

onset of the Second World War and a great increase in the manufacture of radio and [electronic devices](#) ment, developers and manufacturers needed use specific component values for their designs rather than licity of variations available from different component manufacturers.

impetus came about after the Second World War with the introduction and the significant rise in the use of electronic devices and equipment.

the demand for the standardisation needed, an organisation known as the International Electrotechnical ion started work on a standard in 1948. The first release of their document was in 1952 and then this was ited to become document IEC 60063:nnnn, where nnnn is the date of the latest release.

stor E-series and standard values infographic

useful to have a summary infographic about the resistor standard values and E-series. iced one as part of my series of infographics and it summarises the main series in a concise and handy

Resistor E Series

What is the E-series

Resistor (and other component) values are organized into a set of preferred values or standard resistor values known as the E-series.

These standard resistor values have a logarithmically based sequence and this enables the different values to be spaced in such a way that they relate to the component tolerance or accuracy.

Resistor tolerances are generally ±20%, ±10%, ±5%, ±2% and ±1%. More accurate tolerances are available for some resistors, but these are not as widely available and costs are higher.

By having these standard resistor values, electronic components from a variety of manufacturers can be chosen, making sourcing much easier and the cost of the components much less.

E-Series Properties

E SERIES	TOLERANCE (see Note)	NUMBER OF VALUES IN EACH DECADE
E3	±20%	3
E6	±10%	6
E12	±5%	12
E24	±2%	24
E48	±1%	48
E96	±0.5%, ±0.25% and higher tolerances	96

E3, E6, E12 & E24 Series Values

E3 STANDARD RESISTOR SERIES

1.0	3.3	4.7
-----	-----	-----

E6 STANDARD RESISTOR SERIES

1.0	1.5	2.2	3.3	4.7	6.8
-----	-----	-----	-----	-----	-----

E12 STANDARD RESISTOR SERIES

1.0	1.2	1.5	1.8	2.2	2.7	3.3	3.9	4.7	5.6	6.8	8.2
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E24 STANDARD RESISTOR SERIES

1.0	1.1	1.2	1.3	1.5	1.6	1.8	2.0	2.2	2.4	2.7	3.0	3.3	3.6	3.9	4.3	4.7	5.1	5.6	6.2	6.8	7.5	8.2	9.1
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For E48 and E96 values please visit my website: https://www.electronics-notes.com/articles/electronic_components/standard-resistor-values/

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Resistor standard values and E-series

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series resistor values are used universally and provide a very useful selection of resistors to meet the needs of any situation. The series is also used as the basis for other electronic components including capacitors, inductors, and diodes, etc

All types use the E series values, both SMD resistors as well as leaded ones. In fact the E series is used for electronic components like resistors capacitors and inductors whether leaded or surface mount devices.

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Written by **Ian Poole** .

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